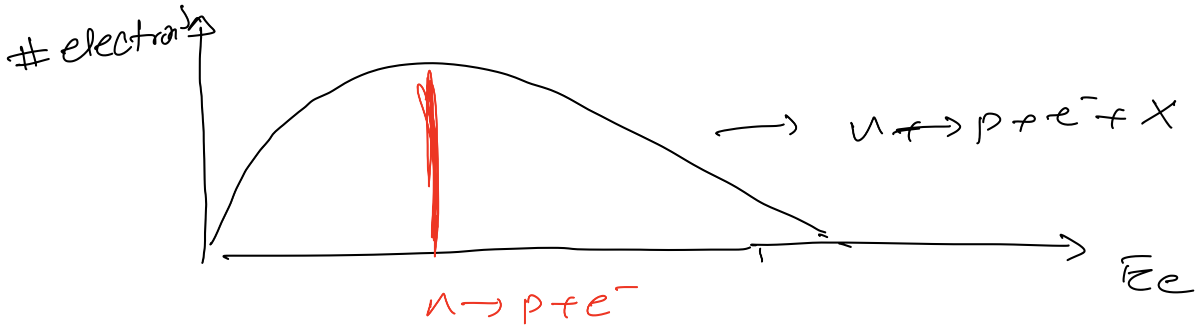
Weak Interactions

First indication β^- decays $\quad n \to p\, e^- \bar{\nu}_e$



X is ν or $\bar{\nu}$?

Reines-Cowen

$n \to p e^- X$

$X \longrightarrow \quad X + p \longrightarrow n + e^+ \quad$ many events.

$n + e^- \quad$ no events

$n + \mu^+ \quad$ no events.

$X = \bar{\nu}_e \implies \nu_e \neq \bar{\nu}_e$

$${}^{A}_{Z}X \to {}^{A}_{Z+1}Y + e^- + \bar{\nu}_e \qquad \beta^-$$

$${}^{A}_{Z}X \to {}^{A}_{Z-1}Y + e^+ + \nu_e \qquad \beta^+$$

1) How to prove $\nu_e \neq \nu_\mu \neq \nu_\tau$

2) $\nu_e \neq \bar{\nu}_e$

$\mathbb{C} \quad n \to \bar{n} \qquad n \neq \bar{n}$

$\mathbb{C} \quad \pi^0 \to \pi^0$

$\mathbb{C} \quad \gamma \to \gamma$

$\mathbb{C} \quad \nu \to \bar{\nu} \quad ? \qquad$ Dirac neutrino

$\mathbb{C} \quad \nu \to \nu \qquad \nu \equiv \bar{\nu} \quad$ Majorana neutrino

open question still to be measured exp.

let's assume $\nu_e \neq \bar{\nu}_e$ $\qquad \begin{pmatrix} \ell^- \\ \nu_e \end{pmatrix} L_\ell = -1 \quad \begin{pmatrix} \ell^+ \\ \bar{\nu}_e \end{pmatrix} L_\ell = +1$

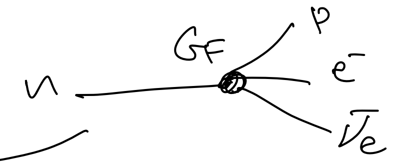
Exp. proof that $\nu_\mu \neq \nu_e$ Exp AGS @ Brookhaven

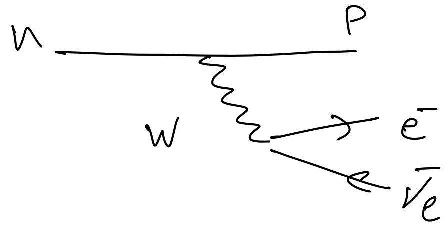
$p + Be \to X$

1962: Lederman – Schwartz – Steinberger Nobel prize 1988

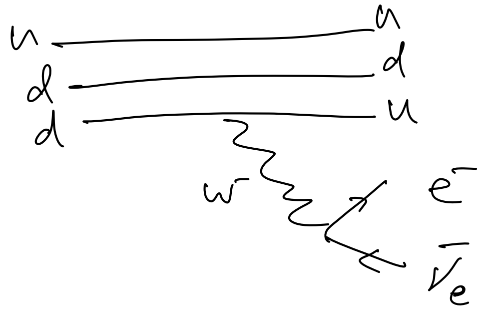
$n \to p + e^- + \bar{\nu}_e \quad Q \geq 0$

$p + \mu^- + \bar{\nu}_\mu \quad Q < 0$





$\longrightarrow$



Muon decay

$\mu^- \to e^- \bar{\nu}_e \nu_\mu$

produces both $\bar{\nu}_e$ and ν_μ.

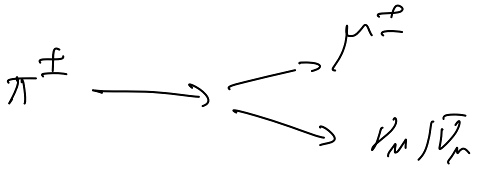
$\Rightarrow$ not useful to distinguish ν_μ from ν_e

Pion decay

$\pi^+ \to \mu^+ \nu_\mu$

$\pi^- \to \mu^- \bar{\nu}_\mu$

$p + Be \to$ many $\pi^\pm + X$



$p + Be \to \pi^+ + X$

$\hookrightarrow \mu^+ \nu_\mu \quad \xrightarrow{\nu_\mu} \quad \nu_\mu + X \to \mu^- + X'$

beam of ν_μ against target : count # events $\begin{matrix} \mu^- + X \\ e^- + X \end{matrix}$

Pions produced

Spark Chamber to detect neutrinos

15 MeV protons

p + Be

$p + p \to p + p + \pi^+ + \pi^-$

3.5×10^{17} protons on target

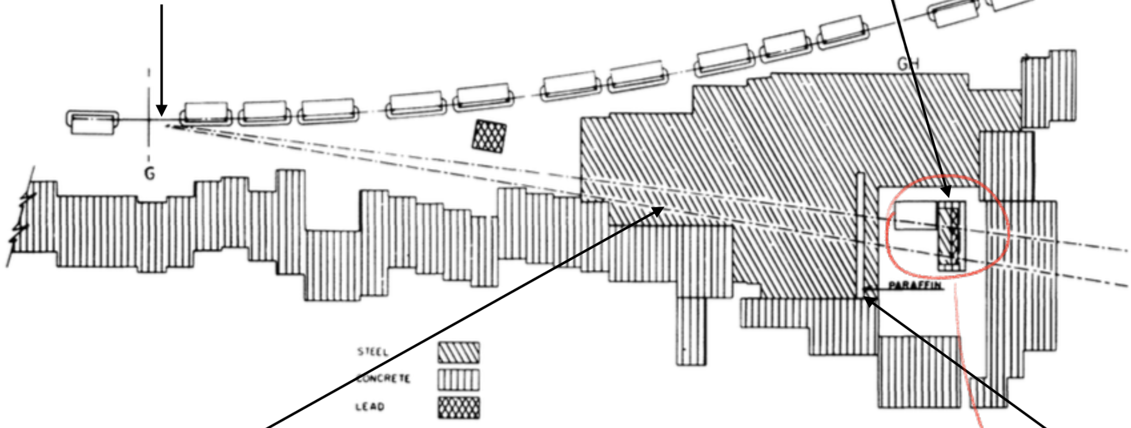
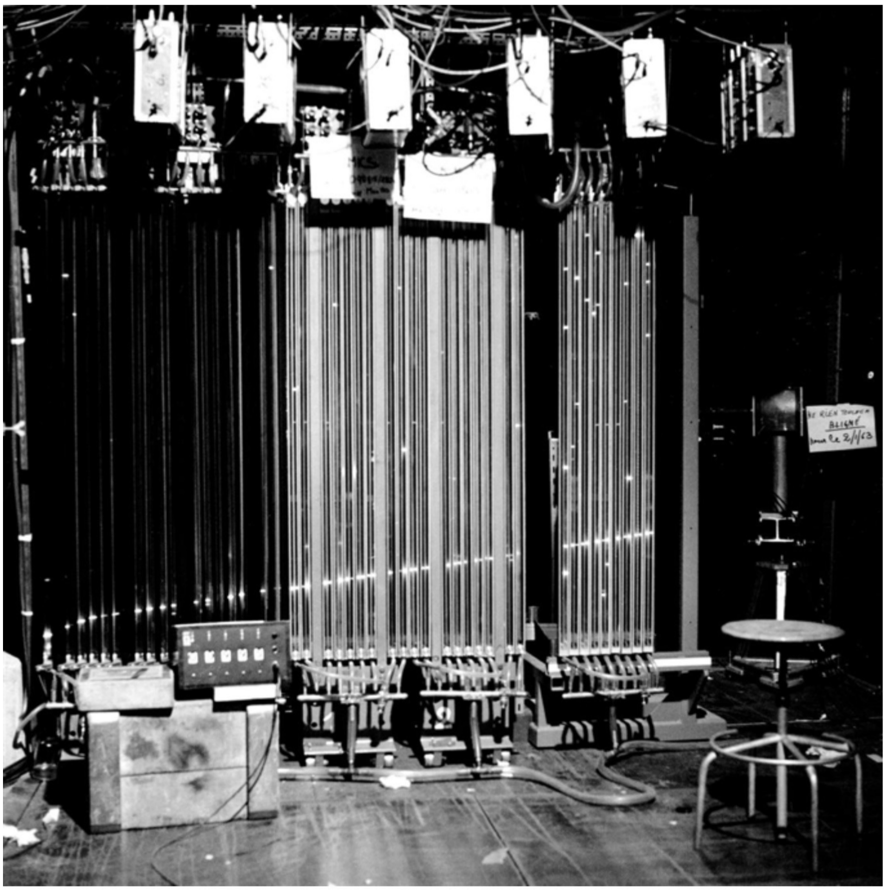


FIG. 1. Plan view of AGS neutrino experiment.

Steel shield stops strongly interacting particles

10 modules of 1T detectors

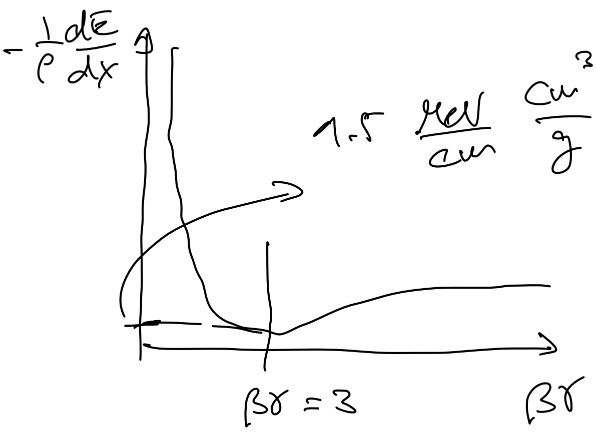
$$p + p \to \pi^{\pm} + X$$
$$\hookrightarrow \mu^{\pm} + \nu_\mu / \bar{\nu}_\mu$$
$$\hookrightarrow \nu_\mu + \bar{\nu}_\mu + X \to \mu^{\pm} + X, \quad e^{\pm} + X$$



Bremsstrahlung for $e^{\pm}$

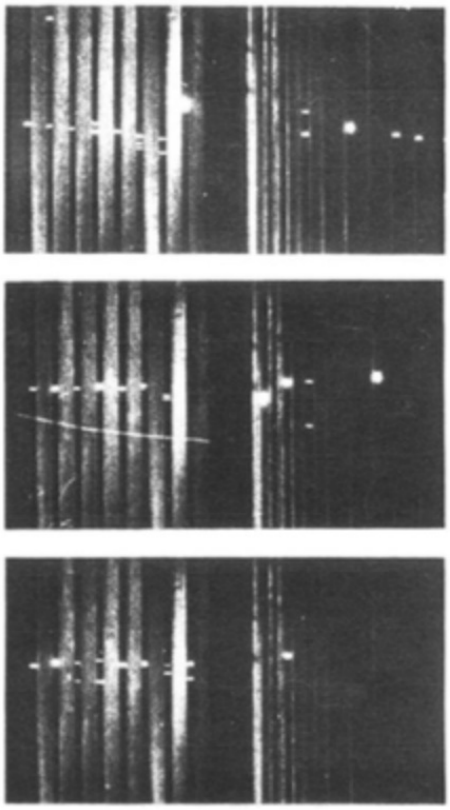
$$E_e > E_c = \frac{600\,\mathrm{MeV}}{Z}$$

Ionization.



$\mu^{\pm}$: tracks only from ionization

$e^{\pm}$: EM showers

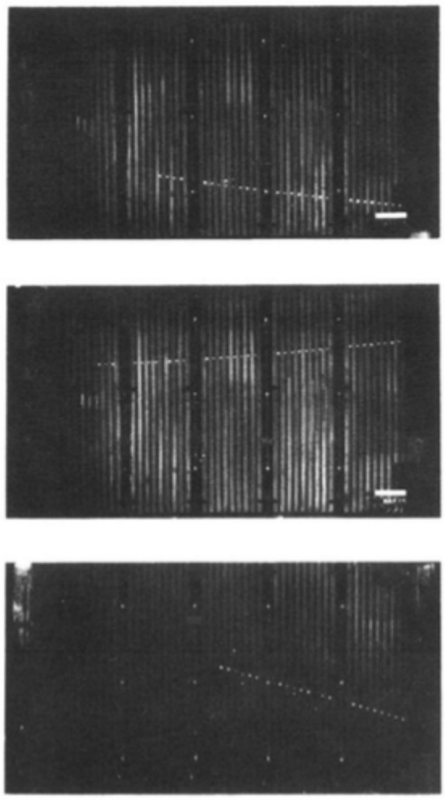


A

B

C

FIG. 8. 400-MeV electrons from the Cosmotron.



A

B

C

FIG. 5. Single muon events. (A) $p_\mu > 540$ MeV and δ ray indicating direction of motion (neutrino beam incident from left); (B) $p_\mu > 700$ MeV/c; (C) $p_\mu > 440$ with δ ray.

μ^- + X : 34 events

6 showers in agreement with expectation. (Background)

$\pi^\pm \to \mu^\pm \, \nu_\mu / \bar{\nu}_\mu$

$\quad\quad e^\pm \, \nu_e / \bar{\nu}_e \simeq 10^{-4}$

$\Rightarrow \quad \nu_\mu \neq \nu_e \qquad \begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}$

ν_τ: 1998 DONUT exp @ Fermilab

Phenomenology of weak interactions

$n \to p \, e^- \bar{\nu}_e$ semileptonic decay of hadrons

$\pi^+ \to \mu^+ \nu_\mu$ leptonic decay of hadrons

$\mu^+ \to \bar{\nu}_\mu \nu_e e^+$ leptonic decay of μ

always produce neutrinos

$\Rightarrow$ unbalanced events $\Rightarrow$ MET missing transv. energy.

Hadronic decays:

$\Lambda(uds) \longrightarrow p\,\pi^-$

$K^+ \longrightarrow \pi^+\pi^0$

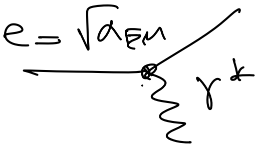
$K^0 \longrightarrow \pi^+\pi^-$

Strangeness $\quad 1 \qquad \emptyset$

τ is large (compared to QCD, QED) $\Rightarrow$ long lifetime.

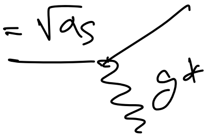
$\Delta S = 1$ only possible in weak decays.

$ie = \sqrt{\alpha_{EM}}$



QED $\qquad \alpha_{EM} = \frac{1}{137}$ $\qquad$ 1 vector boson. $q = \emptyset$

$ig_s = \sqrt{\alpha_s}$



QCD $\qquad \alpha_s = 0.1 \quad (Q^2 \simeq M_Z^2)$

8 vector bosons, $q = \emptyset$
colored

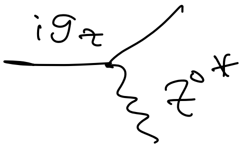
Weak Interactions

ig_W

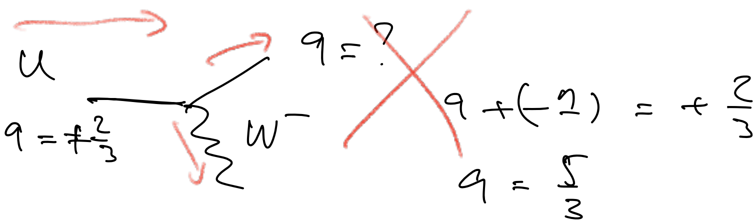
2 charged vector bosons $W^\pm$

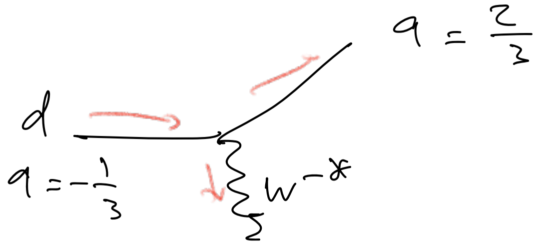
1 neutral vector boson Z^0

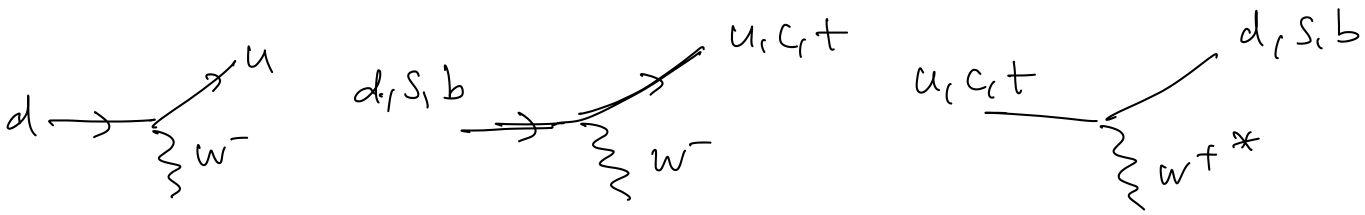
ig_Z



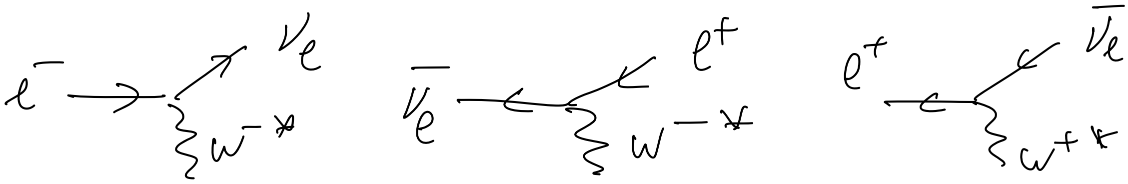
$\alpha_{weak} = \frac{g_W^2}{4\pi} = ?$



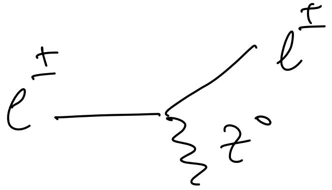


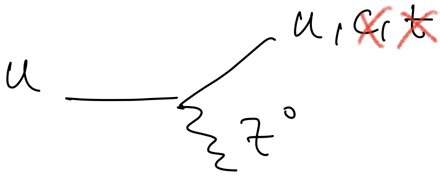


Flavor Changing Charge Current $\qquad \Delta S = 1$, $\Delta C = 1$, $\Delta b = 1$

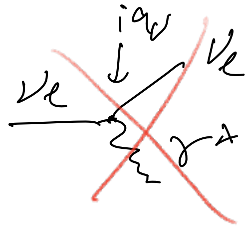


Charged weak current





No Flavor Changing neutral current

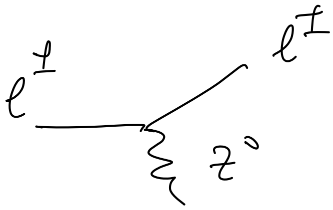


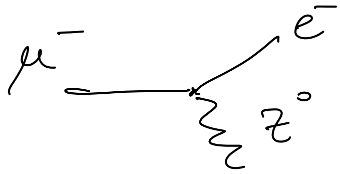
$q_{\nu_e} = \phi$

No electric Charge

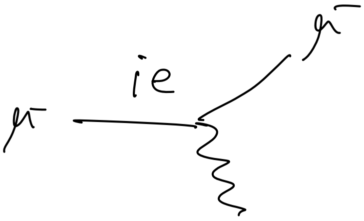
⇓

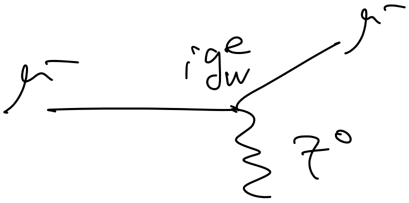
No QED





Not expected is SM.
lepton flavor violation.



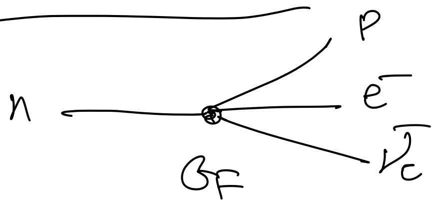


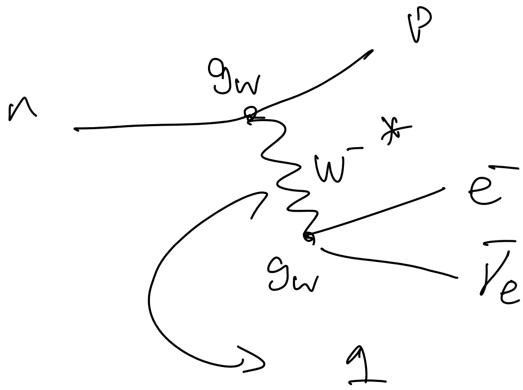
$$\alpha_{EM} = \frac{e^2}{4\pi}$$

$$\alpha_{weak} = \frac{g_w^{e\,2}}{4\pi}$$

$M_W = 80$ GeV
$M_Z = 90$ GeV

Beta decay





$$\frac{1}{q^2 - M_W^2}$$

$Q = m_n - m_p - m_e$
$\simeq 0.5$ MeV

4-fermion Fermi theory.

↓

$\mathcal{M} \sim G_F$

⇓

$\#(n \to p e \bar{\nu}) \propto G_F^2$

⇓ Experiments.

$G_F = 1.16 \times 10^{-5}$ GeV^{-2}
$M_W = 80$ GeV
} measured.

$$\mathcal{M} \sim \frac{g_W^2}{q^2 - M_W^2}$$

$$q^2 \ll M_W^2$$

$$\mathcal{M} \sim \frac{g_W^2}{M_W^2}$$

⇓

$$\#\text{events} \sim \left(\frac{g_W^2}{M_W^2}\right)^2$$

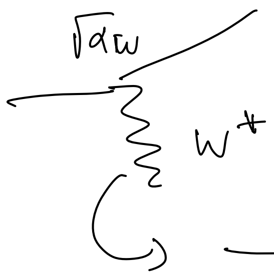
$$G_F = \frac{\sqrt{2}}{8}\frac{g_W^2}{m_W^2} \Rightarrow g_W = 0.653 \Rightarrow \alpha_W = \frac{g_W^2}{4\pi} = \frac{1}{29.5}$$

$$\alpha_S = 0.1 = \frac{1}{11}$$

$$\alpha_{EM} = \frac{1}{137}.$$

$$\alpha_W = \frac{1}{29.5} \qquad \alpha_W > \alpha_{EM}.$$

$$\frac{1}{q^2}$$

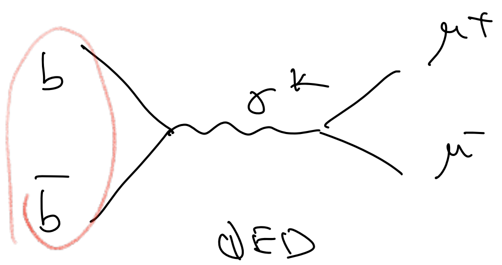


$$\frac{1}{q^2 - M_W^2}$$

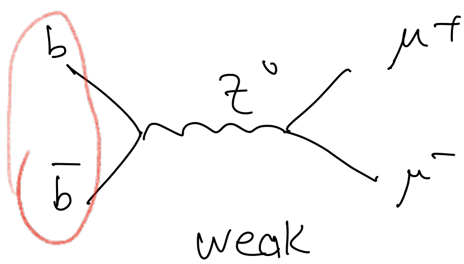
$$M \sim \frac{\alpha_{EM}}{q^2} \qquad M \sim \frac{\alpha_W}{q^2 - M_W^2}$$

$q^2 \ll M_W^2 \Rightarrow$ QED stronger than weak.

$$\Upsilon(b\bar{b}) \longrightarrow \mu^+\mu^-$$



QED



weak

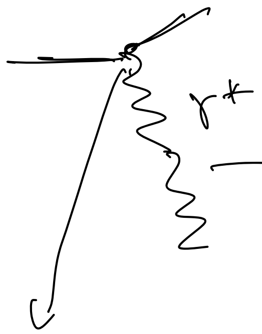
$$q^2 \simeq M_\Upsilon^2 = 100\ \text{GeV}^2 = (10\ \text{GeV})^2$$

$$M_{QED} \sim \frac{\alpha_{EM}}{q^2} \qquad M_{weak} \sim \frac{\alpha_{weak}}{q^2 - M_Z^2} \simeq \frac{\alpha_{weak}}{M_Z^2}$$

negligible.

weak interaction vertex

QED



$\longrightarrow \dfrac{-i\, g_{\mu\nu}}{q^2}$

$ie\gamma^\mu$

$\dfrac{-i(g^{\mu\nu} - q^\mu q^\nu / M_W^2)}{q^2 - M_W^2}$

ig_W?

what is the Lorentz structure of weak Lagrangian?

Exp. properties of weak interactions.

	$\mathbb{P}$	$\mathbb{C}$	$\mathbb{CP}$	(conservation)
QED	✓	✓	✓	
QCD	✓	✓	✓	
weak	✗	✗	✗	

$\mathbb{CP}$ ✗ $\longrightarrow$ Cronin-Fitch exp. $K^0 \to \pi^+\pi^-$, $\pi^+\pi^-\pi^0$

$\mathbb{P}$ ✗ $\longrightarrow$ Wu experiment 1957 $\quad \mathbb{P}$ violated.

$\mathbb{C}$ ✗ $\longrightarrow$ Goldhaber experiment

ν_e left headed $\quad \overset{\Leftarrow}{\longrightarrow} \vec{p} \quad$ LH

$\bar{\nu}_e \quad \overset{\Rightarrow}{\longrightarrow} p \quad$ RH

$\cancel{\mathbb{P}} + \cancel{\mathbb{C}} \Rightarrow$ weak interactions take into account chirality of particles.

For any particle: $\quad$ helicity $h = \dfrac{\vec{p}\cdot\vec{S}}{|\vec{p}|\cdot|\vec{S}|}$ (headedness)

LH: $h = -1 \quad \overset{\overset{\vec{S}}{\Leftarrow}}{\longrightarrow} \vec{p}$

RH: $h = +1 \quad \overset{\Rightarrow}{\longrightarrow}$

For massive particles h is not Lorentz inv. $\Rightarrow$ not conserved

For $m=0$ helicity is Lorentz inv. $\Rightarrow$ prop. of the particle.

$\beta = \frac{p}{E} = \frac{p^2}{\sqrt{p^2+m^2}} \longrightarrow 1$ helicity becomes almost inv. also for massive particles.

Fermion Spinor:

$$\Psi = \Psi_{LH} + \Psi_{RH} = \frac{1}{2}(1-\gamma^5)\Psi + \frac{1}{2}(1+\gamma^5)\Psi$$

$$\gamma^5 \Psi_{LH} = \gamma^5 \frac{1}{2}(1-\gamma^5)\Psi = \frac{1}{2}(\gamma^5 - (\gamma^5)^2)\Psi$$

$$= \frac{1}{2}(\gamma^5 - \mathbb{1})\Psi = -\Psi_{LH}$$

$\gamma^5 \Psi_{RH} = \Psi_{RH}$ LH, RH: Chiral States.

For $\beta \longrightarrow 1$ helicity $\equiv$ Chirality